

Ayesha Dorani

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EDUCATION

King's College London

London, United Kingdom

BSc Computer Science

Sep 2020 – Jun 2026

- **First-Class Honours:** Graduated with a top-tier overall cumulative mark of **81%**.
- **Dissertation (85%):** Engineered a responsive React/TypeScript messaging prototype implementing real-time state synchronisation, pre-send undo mechanisms, and transparent post-send message revocation handles.
- **Amazon Women in Innovation Award - Winner:** Selected for a highly competitive national technology bursary program based on exceptional academic excellence and technical potential.

SKILLS

Programming Languages: Java, Python, JavaScript, TypeScript, SQL, HTML5, CSS3

Technologies: Django, React, JavaFX, Swing, Node.js, Express, MongoDB, REST APIs, JSON

Tools & Concepts: Git, GitHub, OOP (Object-Oriented Programming), Concurrent Programming, System Design, Data Processing

Spoken Languages: English (Native), Pashto (Native)

PROJECTS

Predator–Prey Ecosystem Simulation | *Java, Swing*

- **Built an interactive ecosystem simulation** modelling seven distinct actor species across rainforest and river environments using advanced object-oriented design patterns and polymorphism.
- **Engineered a 120×80 grid-based simulation engine** tracking dynamic breeding parameters, ageing metrics, transmissible disease vectors, and mutation inheritance loops.

Scholar++ | *Python, Django, OpenAlex API*

- **Built a Django academic research platform** integrating OpenAlex data, enabling users to explore multi-stage relationships across authors, works, concepts, venues and institutions.
- **Engineered concurrent API pipelines** via aiohttp and asyncio, implementing robust retry/exponential-backoff workflows for rate limits alongside fast backend filtering, multi-column sorting, and dataset pagination.

Property Viewer | *Java, JavaFX, OpenCSV*

- **Developed a desktop JavaFX property marketplace** for the London rental economy, featuring multi-variable price/date filtering algorithms, interactive geographic vector mapping, and a visual transactional booking workflow.
- **Processed 72,300 live Airbnb records** using OpenCSV parser tools, translating raw dataset properties into clean, structured Java objects to drive instantaneous demographic and borough analytics.

VIRTUAL INTERNSHIPS & TECHNICAL EXPERIENCE

Goldman Sachs

Nov 2024

Software Engineering Virtual Experience (Forage)

- **Conducted cryptographic password-security audits** by utilizing password recovery tools to identify outdated hashing algorithms within leaked database simulations. Authored a technical remediation memo outlining mitigation strategies, including cryptographic salting and strict complexity policies.

Skyscanner

Sep 2026

Software Engineering Virtual Experience (Forage)

- **Developed responsive web and mobile** application features using React and Java, integrating Skyscanner's Backpack design system to build custom user interfaces and flight itinerary components.

Hewlett Packard Enterprise

Sep 2026

Software Engineering Virtual Experience (Forage)

- **Developed backend service layers** using Java Spring Boot to construct a RESTful API web service, implementing JSON data parsing and writing targeted unit tests to ensure accurate endpoint execution.

EXPERIENCE

Ernest Jones

Nov 2024 – Sep 2025

Sales Assistant

Bedford, United Kingdom

- **Consistently secured 100% of weekly sales quotas** in a fast-paced retail client environment, while actively outperforming baseline team KPI targets by 25% for luxury product care packages and supplemental product lines.
- **Managed high-value client transactions** and refined conflict-resolution skills by translating custom requirements into high-value product conversions, displaying commercial acumen and reliable stakeholder handling.

Explore Learning

Nov 2019 – Jul 2020

Academic Tutor

Bedford, United Kingdom

- **Orchestrated cross-functional educational sessions** for diverse learner cohorts aged 4–14, translating abstract algorithmic and mathematical principles into accessible learning frameworks to accelerate skill mastery.